

Adjustable Bulldozer Maintenance Platform

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Introduction & Goal

In collaboration with Rio Tinto, our team is designing an adjustable lift for use on D10 and D11 dozers. During maintenance, these dozers often require engine removal, leaving a large hollow space in the center. Currently, mechanics must crawl underneath the dozer and use a step stool on a platform to access essential components. This method raises concerns about both safety and efficiency. To address these issues, Rio Tinto has requested a mechanically robust lift that can fit beneath the dozer, operate within the hollowed-out space, and reach the necessary working height.



Design and Engineering Decisions

Required Specifications	Units	Value
Platform final height	ft	5
Weight capacity	lbs	700
Top platform area	ft x ft	2x3
Time to raise platform	sec	60
Max controller height	ft	5
Grip tape	Binary	-
Swivel wheels	Binary	-
Lockable wheels	Binary	-
No handrails	Binary	-
Max starting height	ft	2
Height adjusting increment	in	1
Max lift weight	lbs	300
Wheel diameter	in	6

The requirements for the lift are that it must fit underneath the bulldozers when fully compressed (be under 2' in height), extend to a height of 5', be adjustable while on the platform, and be safe for users to operate. To the right is a detailed table of all specifications.

The scissor lift is constructed from 1/4" steel and features two hydraulic cylinders actuated by a manual foot pump. This configuration was chosen to ensure a robust design that does not require an external power source to operate.

The lift uses a double-stacked scissor mechanism to maximize its range of motion and achievable height.

3D Model



Results & Capabilities

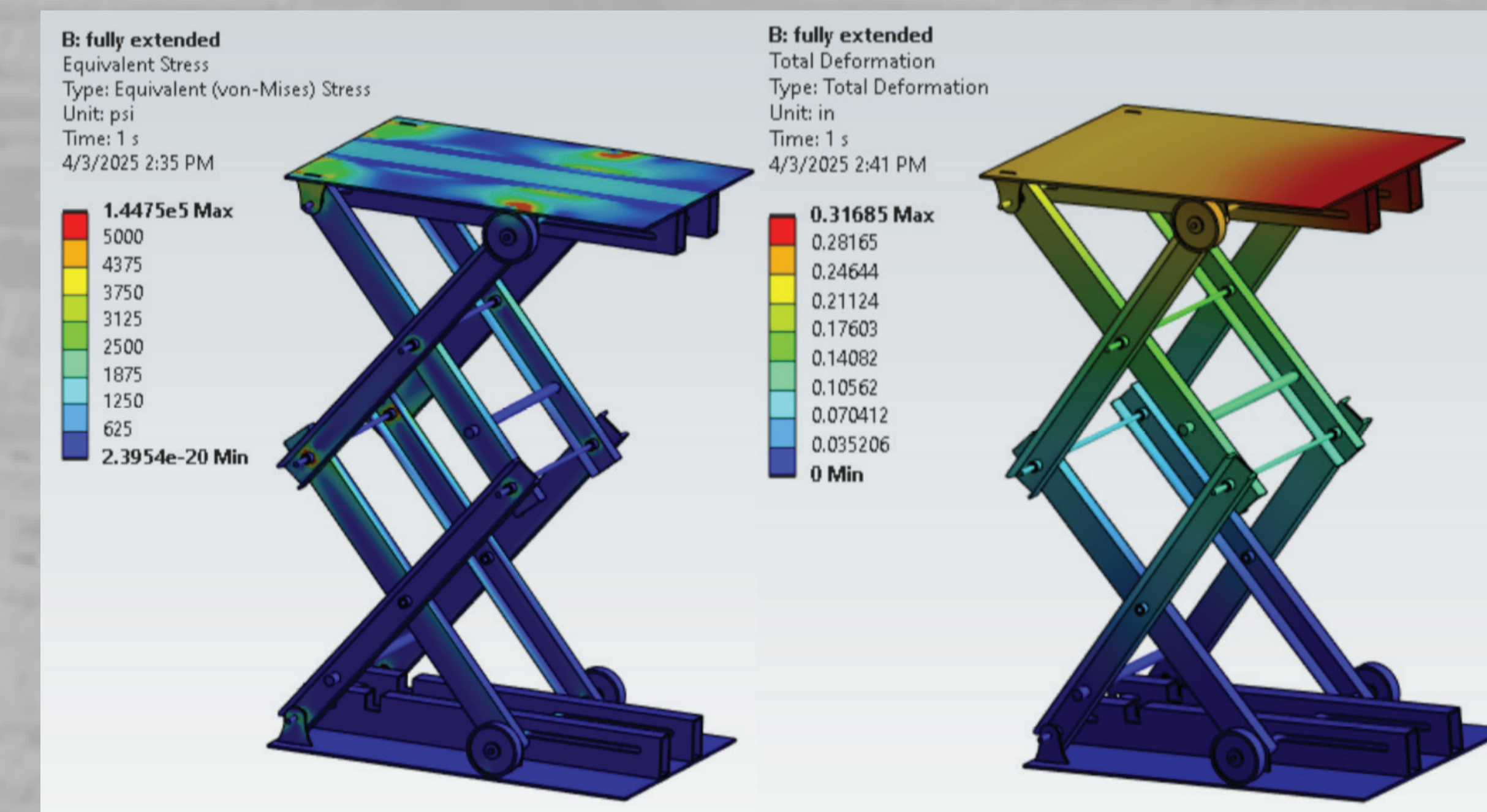
The scissor lift is capable of:

- Lifting users up 5ft.
- Collapsing to a minimum height of 2ft.
- Lifting users with a manual hydraulic pump within 60 seconds.
- Maneuvering easily with 360 degree caster wheels.
- Safe and stable operation.

Analysis

A static structural analysis of the lift assembly was conducted utilizing Ansys Workbench. The objective of the finite element analysis is to evaluate the lift's ability to withstand the maximum weight of the user, 500 lbs, at a fully extended height. This analysis was used to assess the suitability of the selected material, determine the appropriate thickness, and identify areas that experience high stress.

To achieve this, a 500 lb load was applied to the top plate, while the bottom plate was fixed in place. Additionally, individual components of the assembly were meshed independently, to ensure optimal sizing and shape for more accurate results. The maximum Von Mises stress in the linkages was 9000 psi with a maximum deformation of 0.31 inches, giving a factor of safety of 4.



Conclusion

Throughout both semesters, we went through various design iterations, each of which was validated through physical testing and finite element analysis. The first prototype used linkages with self-aligning bearings. However, these components caused instability, prompting further testing on the linkages under lateral forces and additional evaluation of the bearings. As a result, we switched to c-channels and weld-on pivot bushings, which significantly improved the lift's stability and durability. The integration of a hydraulic system into our machined lift resulted in a fully operational platform, which will be delivered to RioTinto.